# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** $3x + 4$ | Linear means degree exactly 1. |
| 2 | **(a)** -10 | Put $x$ in $p(x) = 0$. |
| 3 | **(b)** -1 | Substitute $x = -k$. |
| 4 | **(d)** 7 | Put $x = 2$. |
| 5 | **(a)** $(4, 0)$ | Put $y = 0$. |
| 6 | **(d)** 7 | Find each zero. |
| 7 | **(d)** $6n - 5$ | Use $a_n = a_1 + (n-1)d$. |
| 8 | **(b)** 250 | Fare = fixed + rate × time. |
| 9 | **(c)** $5 - 2x$ | Linear means degree exactly 1. |
| 10 | **(a)** -12 | Put $x$ in $p(x) = 0$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $3x - 18 = 0$ gives $x = 6$ (1); The graph meets the $x$-axis at $(6, 0)$ (1). | 1+1 |
| 12 | $p(2) = 11$ (1); $p(-2) = -5$ (1). | 1+1 |
| 13 | Slope $= 8$, intercept $= 3$ (1); Cuts $y$-axis at $(0, 3)$ (1). | 1+1 |
| 14 | $5x + 20 = 0$ gives $x = -4$ (1); The graph meets the $x$-axis at $(-4, 0)$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $x = -1$: $y = 1$; $x = 0$: $y = 3$ (1); $x = 1$: $y = 5$; $x = 2$: $y = 7$ (1); Slope $= 2$, $y$-intercept $= 3$ (1). | 1+1+1 |
| 16 | Substitute: $11 = 6 + k$ (1); $k = 5$ (1); Line: $y = 2x + 5$ (1). | 1+1+1 |
| 17 | $p + q = 5x + 10$ (1); Put $5x + 10 = 0$ (1); Zero $x = -2$ (1). | 1+1+1 |
| | **OR:** $x = -1$: $y = -4$; $x = 0$: $y = -1$ (1); $x = 1$: $y = 2$; $x = 2$: $y = 5$ (1); Slope $= 3$, $y$-intercept $= -1$ (1). | 1+1+1 |
| 18 | Substitute: $4 = 6 + k$ (1); $k = -2$ (1); Line: $y = 3x - 2$ (1). | 1+1+1 |
| | **OR:** $p + q = 4x - 4$ (1); Put $4x - 4 = 0$ (1); Zero $x = 1$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $C(x) = 12x + 60$ (1); $C(5) = 120$ (1); $12x + 60 = 168$ (1); $x = 9$ (1); Hence the number of notebooks is 9 (1). | 1+1+1+1+1 |
| | **OR:** $y(0) = -12$, $y(1) = -8$, $y(2) = -4$ (1); Crosses $x$-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= 4$ (1); Slope is positive: the graph rises (1). | 1+1+1+1+1 |
| 20 | $C(x) = 15x + 45$ (1); $C(4) = 105$ (1); $15x + 45 = 150$ (1); $x = 7$ (1); Hence the number of notebooks is 7 (1). | 1+1+1+1+1 |
| | **OR:** $y(0) = 9$, $y(1) = 6$, $y(2) = 3$ (1); Crosses $x$-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= -3$ (1); Slope is negative: the graph falls (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** -1 | Substitute $x = -k$. |
| 2 | **(c)** 6 | Put $x = 2$. |
| 3 | **(b)** $(-4, 0)$ | Put $y = 0$. |
| 4 | **(d)** 8 | Find each zero. |
| 5 | **(d)** $3n + 5$ | Use $a_n = a_1 + (n-1)d$. |
| 6 | **(b)** 390 | Fare = fixed + rate × time. |
| 7 | **(b)** $\frac{x}{3} + 1$ | Linear means degree exactly 1. |
| 8 | **(a)** -10 | Put $x$ in $p(x) = 0$. |
| 9 | **(c)** -14 | Substitute $x = -k$. |
| 10 | **(c)** 13 | Put $x = 2$. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $p(4) = 7$ (1); $p(-4) = -17$ (1). | 1+1 |
| 12 | Slope $= -5$, intercept $= 9$ (1); Cuts $y$-axis at $(0, 9)$ (1). | 1+1 |
| 13 | $2x + 14 = 0$ gives $x = -7$ (1); The graph meets the $x$-axis at $(-7, 0)$ (1). | 1+1 |
| 14 | $p(1) = 8$ (1); $p(-1) = -4$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $x = -1$: $y = 7$; $x = 0$: $y = 5$ (1); $x = 1$: $y = 3$; $x = 2$: $y = 1$ (1); Slope $= -2$, $y$-intercept $= 5$ (1). | 1+1+1 |
| 16 | Substitute: $9 = 4 + k$ (1); $k = 5$ (1); Line: $y = 4x + 5$ (1). | 1+1+1 |
| 17 | $p + q = 6x + 6$ (1); Put $6x + 6 = 0$ (1); Zero $x = -1$ (1). | 1+1+1 |
| | **OR:** $x = -1$: $y = -2$; $x = 0$: $y = 2$ (1); $x = 1$: $y = 6$; $x = 2$: $y = 10$ (1); Slope $= 4$, $y$-intercept $= 2$ (1). | 1+1+1 |
| 18 | Substitute: $6 = 15 + k$ (1); $k = -9$ (1); Line: $y = 5x - 9$ (1). | 1+1+1 |
| | **OR:** $p + q = 6x + 6$ (1); Put $6x + 6 = 0$ (1); Zero $x = -1$ (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $C(x) = 20x + 90$ (1); $C(3) = 150$ (1); $20x + 90 = 250$ (1); $x = 8$ (1); Hence the number of notebooks is 8 (1). | 1+1+1+1+1 |
| | **OR:** $y(0) = 10$, $y(1) = 15$, $y(2) = 20$ (1); Crosses $x$-axis: $x = -2$ (1); Point $(-2, 0)$ (1); Slope $= 5$ (1); Slope is positive: the graph rises (1). | 1+1+1+1+1 |
| 20 | $C(x) = 14x + 70$ (1); $C(6) = 154$ (1); $14x + 70 = 210$ (1); $x = 10$ (1); Hence the number of notebooks is 10 (1). | 1+1+1+1+1 |
| | **OR:** $y(0) = -8$, $y(1) = -10$, $y(2) = -12$ (1); Crosses $x$-axis: $x = -4$ (1); Point $(-4, 0)$ (1); Slope $= -2$ (1); Slope is negative: the graph falls (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** $(5, 0)$ | Put $y = 0$. |
| 2 | **(b)** 10 | Find each zero. |
| 3 | **(a)** $9n - 6$ | Use $a_n = a_1 + (n-1)d$. |
| 4 | **(b)** 280 | Fare = fixed + rate × time. |
| 5 | **(b)** $4x$ | Linear means degree exactly 1. |
| 6 | **(b)** -24 | Put $x$ in $p(x) = 0$. |
| 7 | **(a)** -11 | Substitute $x = -k$. |
| 8 | **(c)** 5 | Put $x = 2$. |
| 9 | **(d)** $(-4, 0)$ | Put $y = 0$. |
| 10 | **(b)** 13 | Find each zero. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Slope $= 9$, intercept $= -2$ (1); Cuts $y$-axis at $(0, -2)$ (1). | 1+1 |
| 12 | $7x - 28 = 0$ gives $x = 4$ (1); The graph meets the $x$-axis at $(4, 0)$ (1). | 1+1 |
| 13 | $p(3) = 22$ (1); $p(-3) = -8$ (1). | 1+1 |
| 14 | Slope $= -7$, intercept $= -4$ (1); Cuts $y$-axis at $(0, -4)$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $x = -1$: $y = -8$; $x = 0$: $y = -3$ (1); $x = 1$: $y = 2$; $x = 2$: $y = 7$ (1); Slope $= 5$, $y$-intercept $= -3$ (1). | 1+1+1 |
| 16 | Substitute: $3 = 10 + k$ (1); $k = -7$ (1); Line: $y = 2x - 7$ (1). | 1+1+1 |
| 17 | $p + q = 6x - 6$ (1); Put $6x - 6 = 0$ (1); Zero $x = 1$ (1). | 1+1+1 |
| | **OR:** $x = -1$: $y = 9$; $x = 0$: $y = 6$ (1); $x = 1$: $y = 3$; $x = 2$: $y = 0$ (1); Slope $= -3$, $y$-intercept $= 6$ (1). | 1+1+1 |
| 18 | Substitute: $20 = 12 + k$ (1); $k = 8$ (1); Line: $y = 3x + 8$ (1). | 1+1+1 |
| | **OR:** $p + q = 5x - 10$ (1); Put $5x - 10 = 0$ (1); Zero $x = 2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $C(x) = 11x + 55$ (1); $C(7) = 132$ (1); $11x + 55 = 121$ (1); $x = 6$ (1); Hence the number of notebooks is 6 (1). | 1+1+1+1+1 |
| | **OR:** $y(0) = -18$, $y(1) = -12$, $y(2) = -6$ (1); Crosses $x$-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= 6$ (1); Slope is positive: the graph rises (1). | 1+1+1+1+1 |
| 20 | $C(x) = 25x + 120$ (1); $C(2) = 170$ (1); $25x + 120 = 395$ (1); $x = 11$ (1); Hence the number of notebooks is 11 (1). | 1+1+1+1+1 |
| | **OR:** $y(0) = 20$, $y(1) = 16$, $y(2) = 12$ (1); Crosses $x$-axis: $x = 5$ (1); Point $(5, 0)$ (1); Slope $= -4$ (1); Slope is negative: the graph falls (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.